Advanced Statistics

Regression analysis - generalized linear model

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FIRS - Track 1

Limits of the linear model

Some data types are poorly adapted to the standard linear model with Gaussian distribution:

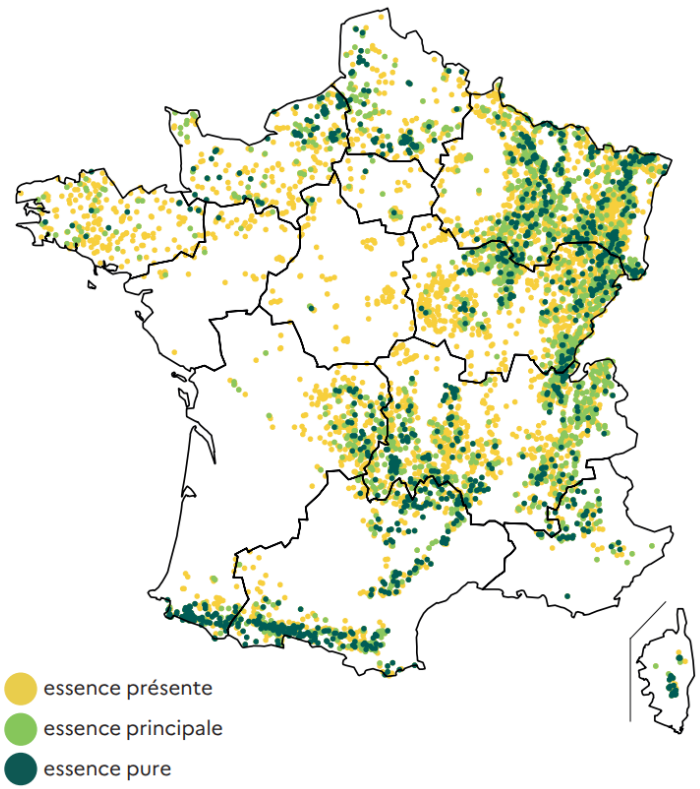
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Limits of the linear model

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- Count data (number of birds)
- Species distribution (presence / absence)



IGN (2023)

Limits of the linear model

Some data types are poorly adapted to the standard linear model with Gaussian distribution:

- Count data (number of birds)
- Species distribution (presence / absence)
- Success in an experimental trial (survival, ...)



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Enters Generalized linear models

- GLM extend the classical linear models by allowing the analyst to use different distributions more suited to the data at hand
- Linear model is a particular case of GLM

A Facelift for the Linear Model

The classical formulation of the linear model is:

$$y = \alpha + \beta * x + \epsilon, \epsilon \sim N(0, \sigma) \tag{1}$$

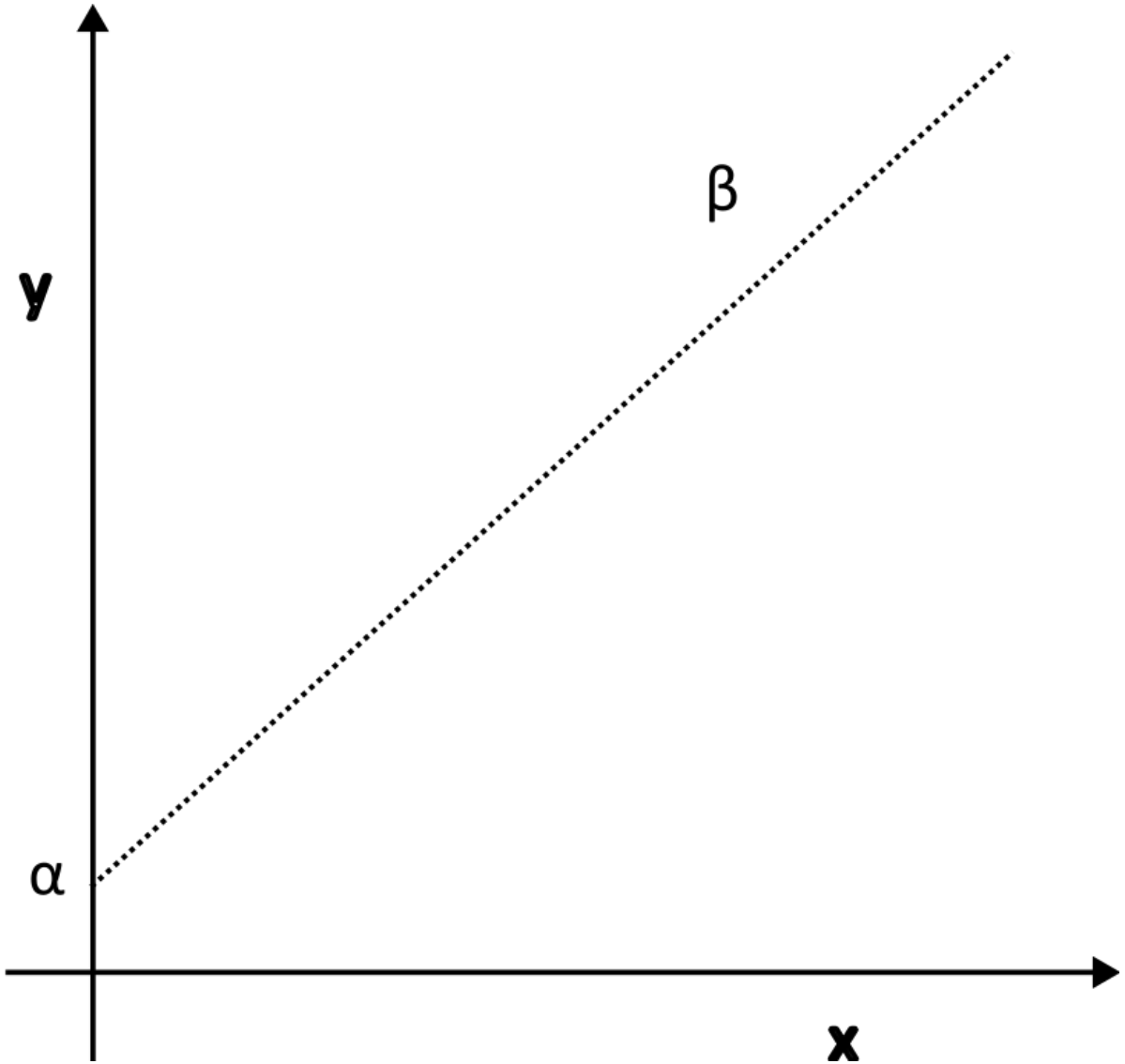
This is equivalent to the following formulation:

$$\begin{aligned} y &\sim N(\mu, \sigma) \\ \mu &= \alpha + \beta * x \end{aligned} \tag{2}$$

In this formulation, μ is the mean value predicted by the model, and σ is the uncertainty around this value (the residual variation).

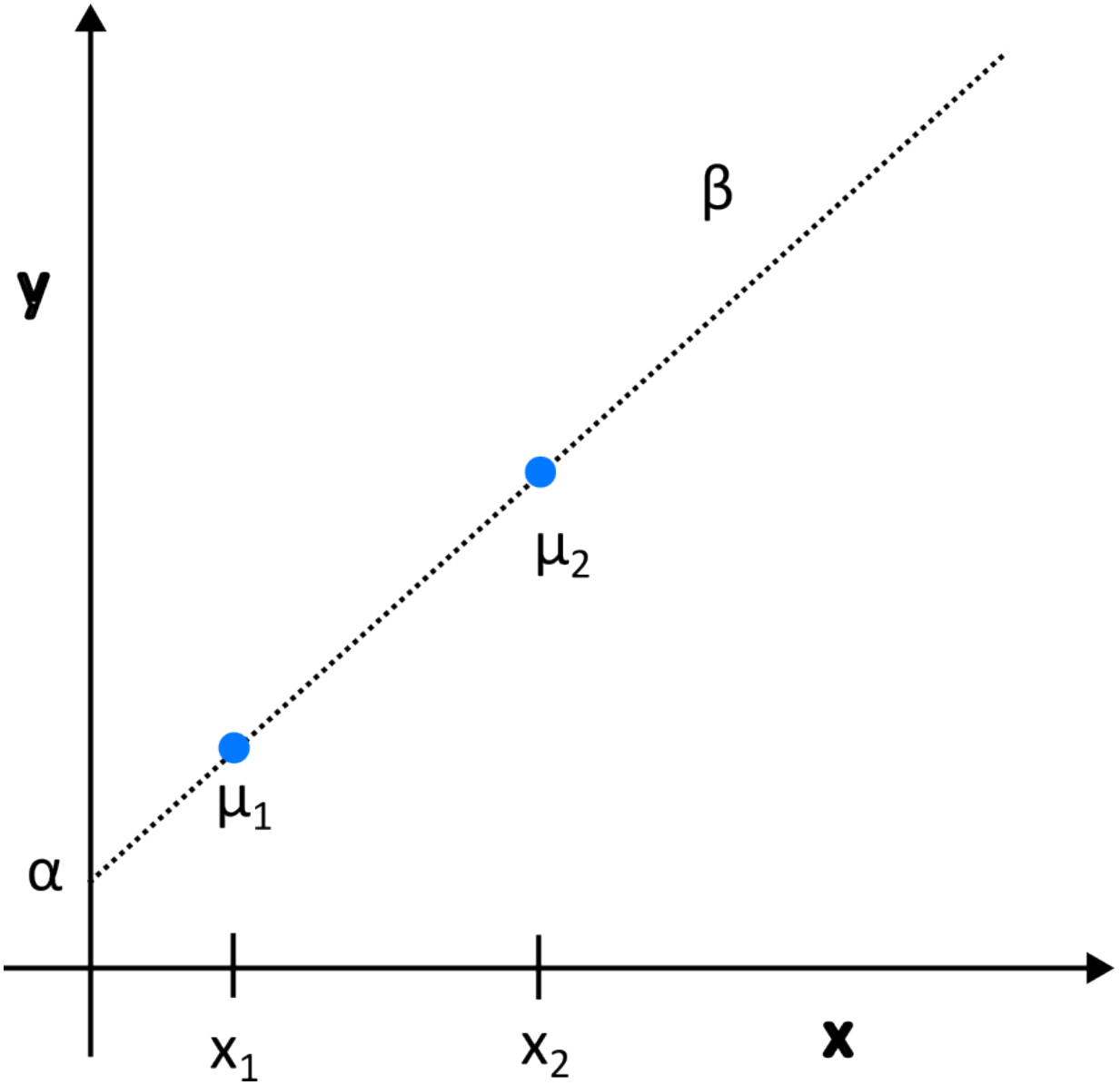
A Facelift for the Linear Model

A graphical representation may help:



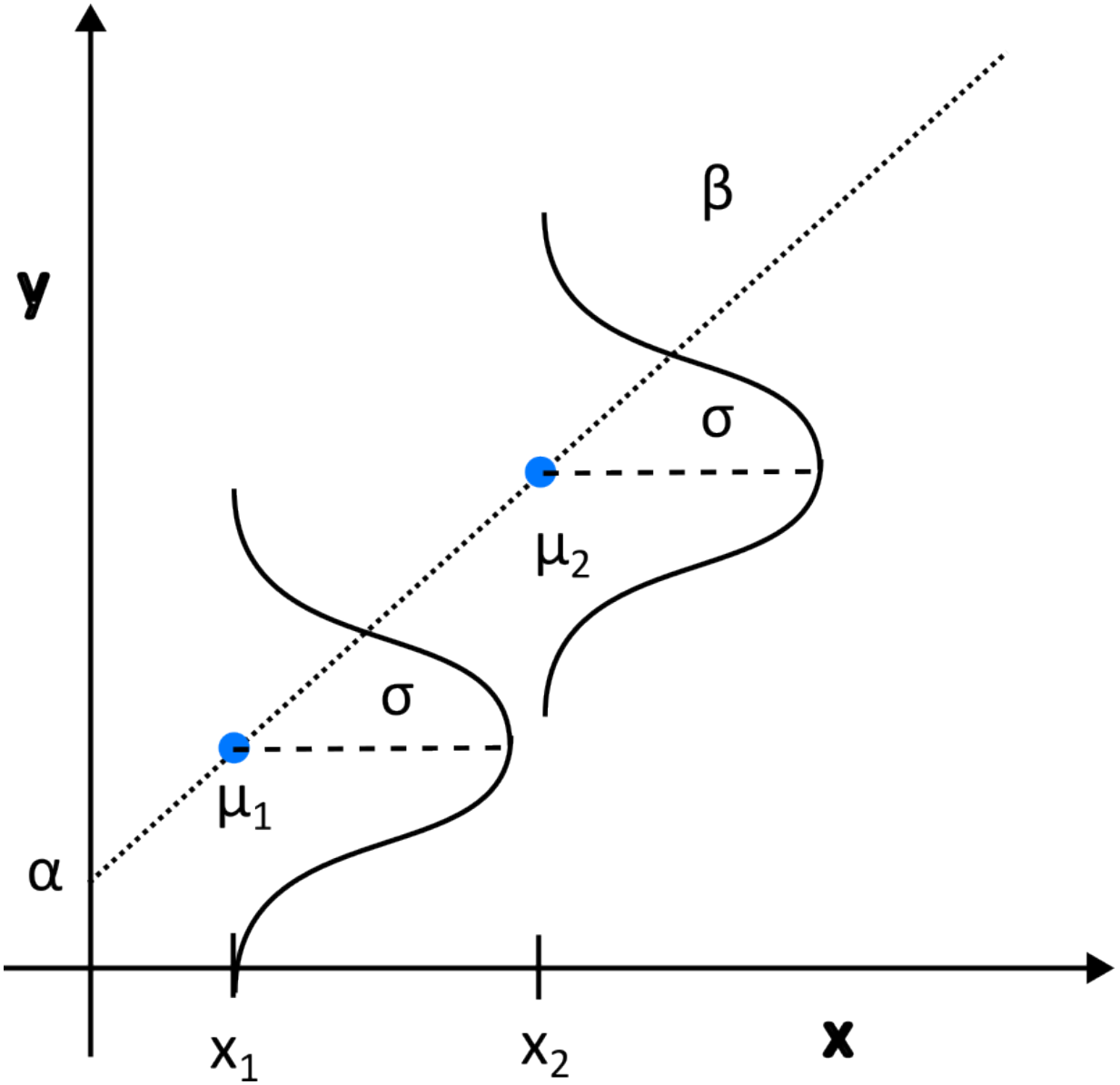
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A Facelift for the Linear Model

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Why This New Formulation Matters

The new formulation of the linear model:

$$\begin{aligned} y_i &\sim N(\mu_i, \sigma) \\ \mu_i &= \alpha + \beta * x_i \end{aligned} \tag{3}$$

Naturally extends to the Generalized linear model such as:
The Poisson distribution for count data

$$\begin{aligned} y_i &\sim P(\lambda_i) \\ f(\lambda_i) &= \alpha + \beta * x_i \end{aligned} \tag{4}$$

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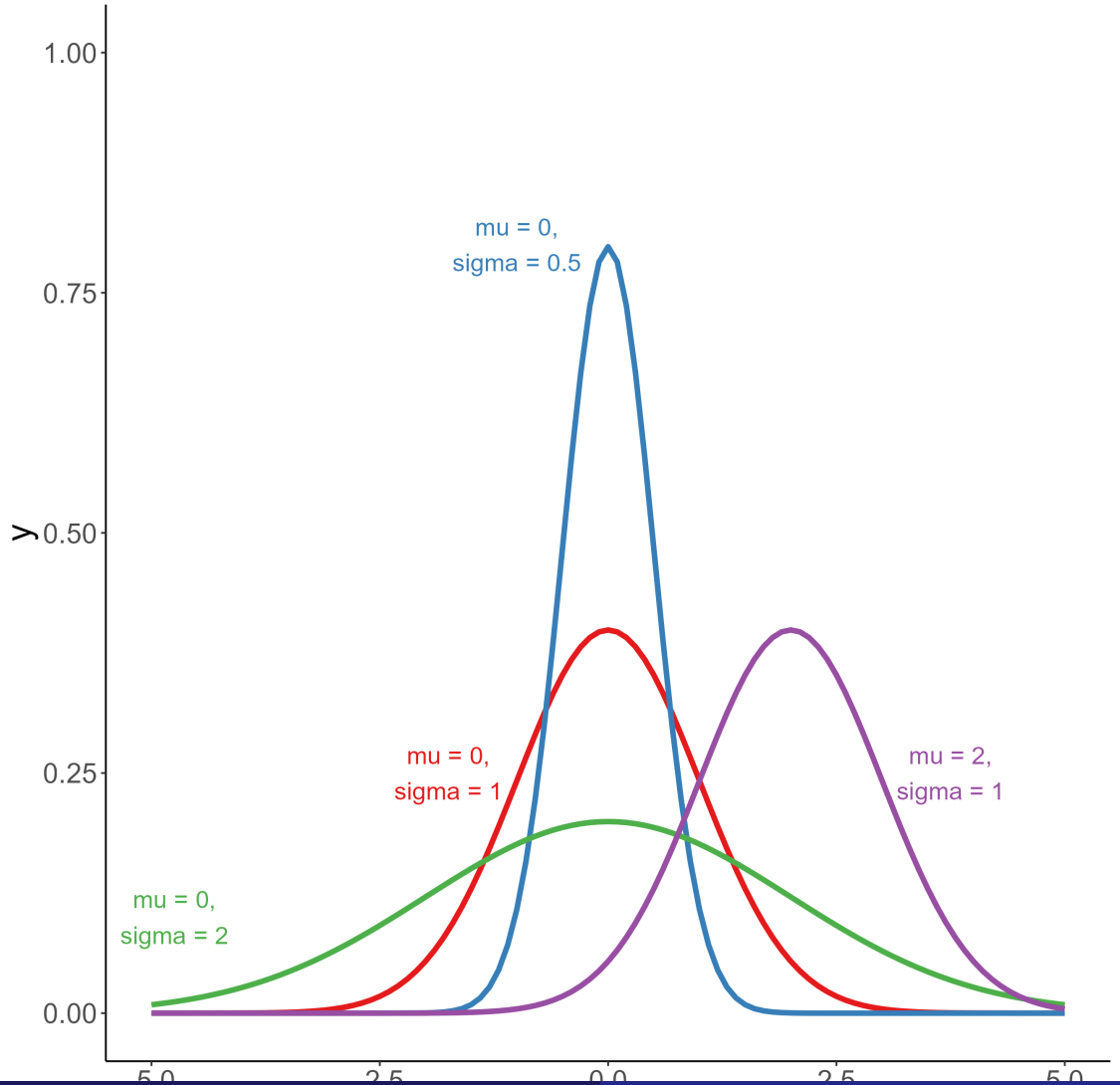
Naturally extends to the Generalized linear model such as:
The Binomial distribution for trial data

$$\begin{aligned} y_i &\sim B(\rho_i, n) \\ f(\rho_i) &= \alpha + \beta * x_i \end{aligned} \tag{4}$$

Statistical Distribution – Normal Distribution

The normal distribution is a **continuous** distribution defined by two parameters:

1. μ: the location of the peak of the distribution, corresponding to its expectation (mean)
2. σ: the spread of the distribution (more precisely, its standard deviation)

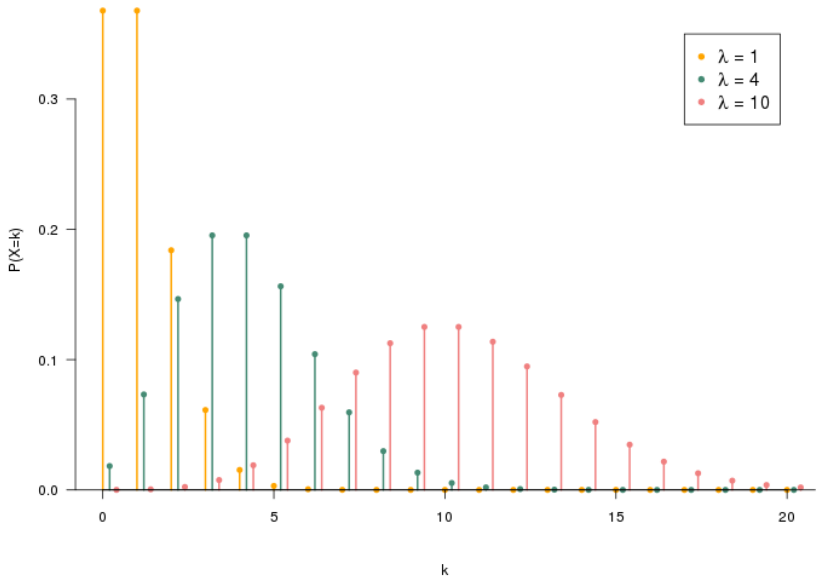


Statistical Distribution – Poisson Distribution

The Poisson distribution is a **discrete** distribution defined by a single parameter:

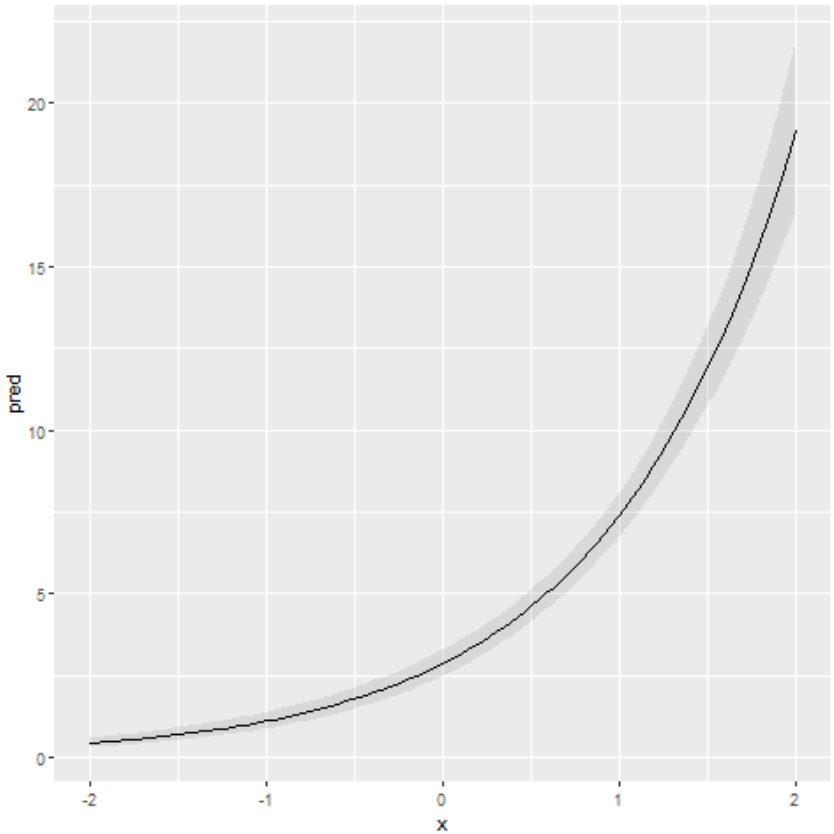
1. λ: determines both the location of the peak (the mean) and the variance of the distribution
2. For a Poisson distribution, mean = variance = λ

Graphically:



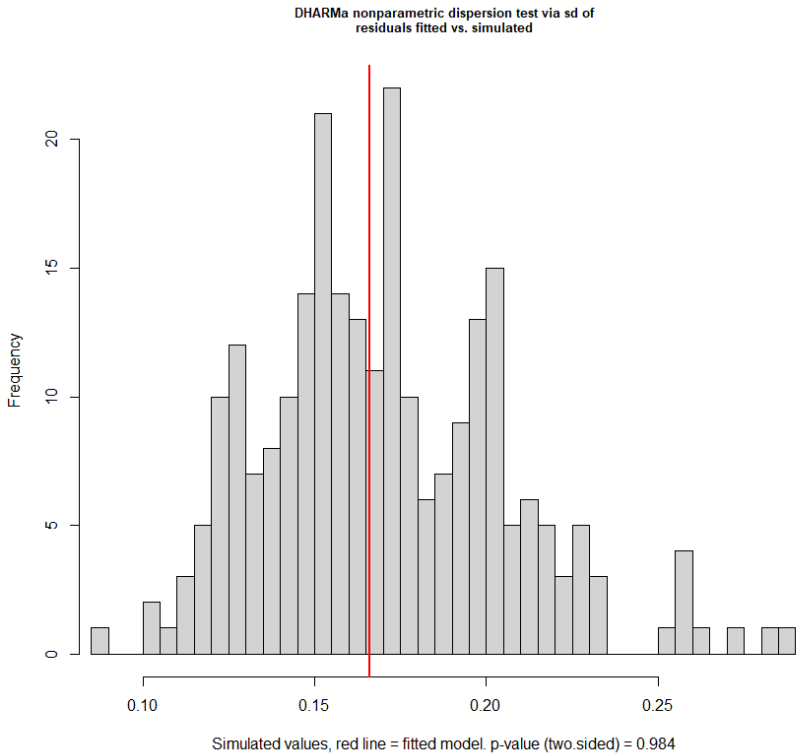
Statistical Distribution – Poisson Distribution

The variance is equal to the mean in a Poisson distribution, implying that observations with larger expected values exhibit greater variability.



Model Diagnostics – Poisson Distribution

This relationship between the variance and the mean should therefore be checked after model fitting to identify potential overdispersion or underdispersion.



The other assumptions to check remain the same as for the linear model: (i) validity, (ii) independence of observations, and (iii) linearity of covariable effects.

The Link Function

Question: Based on your growing knowledge of Poisson regression, what problem could arise with the following model?

$$\begin{aligned} y &\sim P(\lambda) \\ \lambda &= \alpha + \beta * x \end{aligned} \tag{5}$$

The Link Function

The Poisson distribution is only defined for positive values of λ ($\lambda \geq 0$). Therefore, the regression model must ensure that λ never enters the forbidden territory of negative values.
The solution is to use a function *f* that links the regression component ($\alpha + \beta * x$) to λ.
Question: Which of the following functions could work?

1. Reciprocal ($\frac{1}{\alpha+\beta*x}$)
2. Square root ($\sqrt{\alpha + \beta * x}$)
3. Exponential ($e^{\alpha+\beta*x}$)

The Link Function

Several link functions could work. In theory, any function that maps real numbers to strictly positive values is valid.
In practice, the standard choice is the exponential function:

$$\lambda = e^{\alpha+\beta*x} \tag{6}$$

This defines the corresponding link function, which is the natural logarithm:

$$\log(\lambda) = \alpha + \beta * x \tag{7}$$

These two functions (the log link function and its inverse, the exponential function) allow us to move back and forth between the regression component and λ.

The Bernoulli Distribution

Regression models for binary data (0/1) rely on the **Bernoulli** distribution.
This distribution has a single parameter: the probability of "success" (obtaining a 1).

$$Bern(p) \tag{8}$$

The mean of the distribution is p and the variance $p(1-p)$

The Bernoulli Distribution

Question: since **p** is a probability, what are its constraints?

1. p must be greater than or equal to zero
2. p must be less than or equal to one
3. p must be discrete

The Logistic Function

Since **p** is a probability, it must be continuous and lie between 0 and 1.
As with Poisson regression, we therefore need a link function between the regression component $\alpha + \beta * x$ and the parameter p.
This function is the logistic function (fasten your seatbelts):

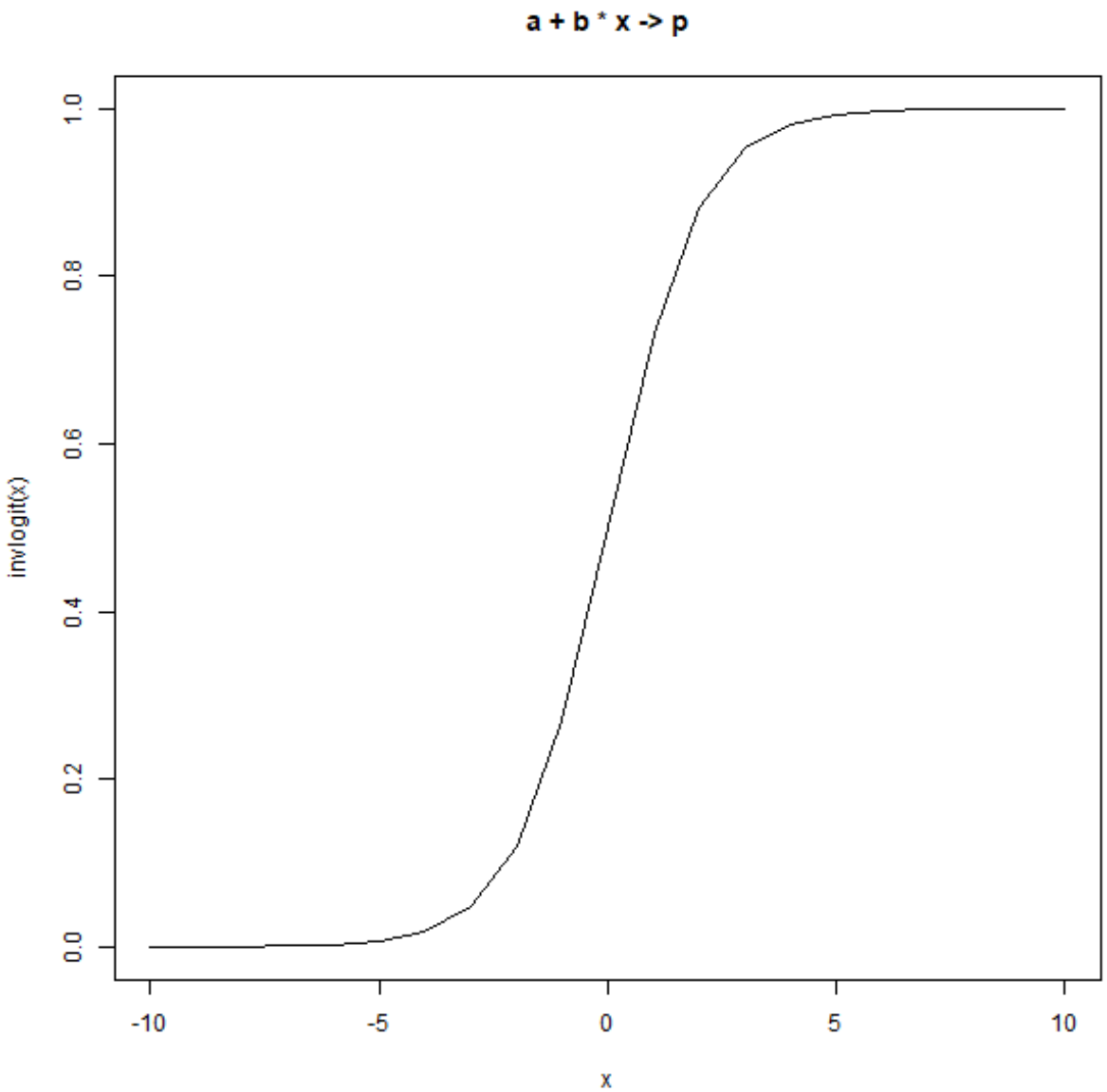
$$logit(x) = log\left(\frac{x}{1-x}\right) \tag{9}$$

And its inverse:

$$logit^{-1}(x) = \frac{e^x}{1+e^x} \tag{10}$$

The Logistic Function

The logistic function:



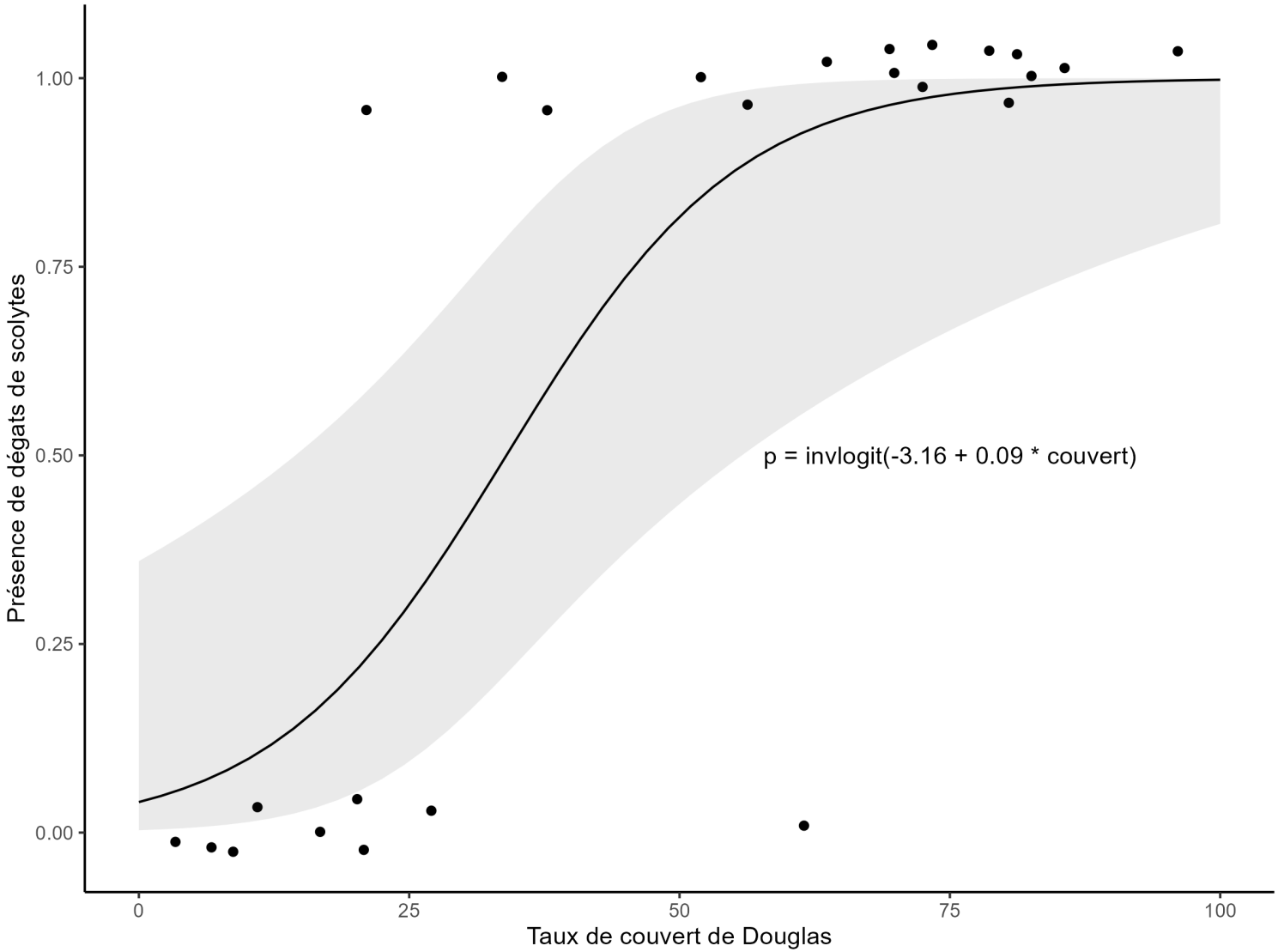
The Logistic Model

The GLM for binary data is also commonly referred to as the **logistic regression model**, after its link function.
It can be written as:

$$\begin{aligned} y &\sim Bern(p) \\ logit(p) &= \alpha + \beta * x \\ p &= logit^{-1}(\alpha + \beta * x) \end{aligned} \tag{11}$$

Logistic Regression in Practice

Presence of bark beetle damage as a function of Douglas-fir canopy cover.



From bernoulli to binomial

Multiple bernoulli trials lead to the binomial distribution.
For exemple, we observe the survival of 10 seedlings in different conditions.
The regression model becomes:

$$survival_i \sim Binomial(n, p_i)$$

With p_i the probaiblity of survival for each seedlings and n the number of Bernoulli trials (here 10). The mean of the binomial distribution is np and its variance $np(1-p)$

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2. The Poisson distribution is defined by a single positive parameter, λ.
3. In a Poisson distribution, the mean and the variance are both equal to λ (spoiler: extensions exist that relax this equality).
4. The link function is the log.

Key Takeaways Bernoulli/Binomial

- The Bernoulli and Binomial distributions are used to model success/failure (0/1) data, such as presence/absence or number of surviving seedlings.

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- These distributions are defined by p, the probability of success and n the number of trials (1 by definition for Bernoulli).
- The mean is np and the variance $np(1-p)$ (spoiler: extensions exist to relax this)
- The link function is the logistic function, hence the name *logistic regression* often used for this model.